

Group Assignment

10 members per group

Our Group Topic: Microservice Architecture and Prototype

Main Role

This group will redesign the system using a microservice architecture.

Work to Be Done

The group should study the current system and identify whether it can be decomposed into independent services. They should define possible microservices such as Student Service, Examination Service, Course Service, Result Service, Transcript Service, Notification Service, Authentication Service, and Reporting Service.

They should explain service boundaries, data ownership, communication methods, API design, deployment approach, and challenges.

The group should also develop a small prototype to demonstrate communication between at least two services.

Expected Deliverables

- Microservice architecture design document
- Service identification table
- Bounded context diagram
- API endpoint design
- Service communication diagram
- Database-per-service or shared database discussion
- Prototype with at least two services
- Benefits and risks of microservices
- AI usage report
 - Report should contains with:
 - AI tools used.
 - Purpose of using each tool.
 - Example prompts used.
 - AI-generated outputs.
 - Human corrections made.
 - Validation method.
 - Mistakes or hallucinations identified.
 - Ethical, privacy, or governance concerns.
 - Reflection on whether AI improved the software engineering process.

Common Evaluation Criteria

The project may be evaluated using the following criteria:

- Understanding of the existing system
- Correct application of Requirements Engineering concepts
- Quality of documentation
- Quality of diagrams and models
- Correct distinction between functional, non-functional, and domain requirements
- Practical use of AI tools
- Human validation of AI outputs
- Creativity and feasibility of the proposed solution
- Quality of prototype, where applicable
- Clarity of presentation and final report